


Problem Sheet 1.

$$1. (a) \quad \nabla \phi = (2x + yz, xz, xy)$$

$$\nabla^2 \phi = \nabla \cdot \nabla \phi = 2 + 0 + 0 = 2$$

$$(b) \quad \nabla \phi = (\phi \cdot k \cot l x, \phi \cdot l \cot k y, \phi \cdot \sqrt{k^2 + l^2})$$

$$\nabla^2 \phi = \nabla \cdot \nabla \phi = -l^2 \phi - k^2 \phi + (k^2 + l^2) \phi = 0$$

$$2. (a) \quad \nabla \cdot u = y + 0 + 0 = y$$

$$\nabla \times u = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy & z^2 & xz \end{vmatrix} = (-1 - 2z, 0 - 1, 0 - x) = (-1 - 2z, -1, -x)$$

$$(b) \quad \nabla \cdot u = 0 + (-z^2) \cos y + 2xz + 2 \cos y$$

$$\nabla \times u = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ u_x & u_y & u_z \end{vmatrix} = \begin{pmatrix} -2z \sinh y + 2yz + 2z \sinh y - 2xy \\ y^2 - y^2 \\ 2yz - 2yz \end{pmatrix} = (0, 0, 0)$$

$$\phi = xy^2z + f(y, z) \quad \frac{\partial f}{\partial y} = -z^2 \sinh y + 2xy^2z - 2xy^2z = -z^2 \sinh y$$

$$f(y, z) = z^2 \cos y + g(z) \quad \frac{\partial g}{\partial z} = 2z \cos y + xy^2 - xy^2 - 2z \cos y = 0$$

$$\phi = xy^2z + z^2 \cos y$$

$$3. \quad \nabla u = (y^2, 2xy + 2z, 2y)$$

$$\nabla u \big|_{(-2, 2, 3)} = (4, -2, 4)$$

$$p: a(x+2) + b(y-2) + (z-3) = 0$$

$$\left(\frac{1}{a}, \frac{1}{b}, -2\right) \times (4, -2, 4) = 0$$

$$a = 2, b = -1$$

$$p: 2x + 4 - y + 2 + z - 3 = 0$$

$$\Leftrightarrow 2x - y + z = -3$$

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$$(a) \quad \nabla \phi = (2x + y - 3, 2y + x, 2z)$$

$$(b) \quad (2, -1, 0)$$

$$(c) \quad \phi_{\min} = 4 + 1 + (-2) - 6 = -3$$

$$5. \quad \nabla \times \phi = \begin{pmatrix} 1 - 1 \\ ax - 2x \\ \cos x - b \cos x \end{pmatrix} = \vec{0}$$

$$a = 2, b = 1$$

$$b. \quad (a) \quad A = (0, 0, xz^2)$$

$$\nabla^2 A = \nabla \cdot \nabla A = (0, 0, 6xz)$$

$$(b) \quad \nabla \times A = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix} = \begin{pmatrix} 0 \\ -z^2 \\ 0 \end{pmatrix}$$

$$\nabla \times (\nabla \times A) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0 & -z^2 & 0 \end{vmatrix} = \begin{pmatrix} 2z^2 \\ 0 \\ 0 \end{pmatrix}$$

$$(c) \quad \nabla \cdot A = 0 + 0 + 3xz^2 = 3xz^2$$

$$\nabla (\nabla \cdot A) = (3z^2, 0, 6xz)$$

$$\nabla \times (\nabla \times A) = \nabla (\nabla \cdot A) - \nabla^2 A$$

Problem Sheet 2

1. (a) $C: x=t \quad y=t^2-2 \quad z=t^3 \quad t \in (0,1)$

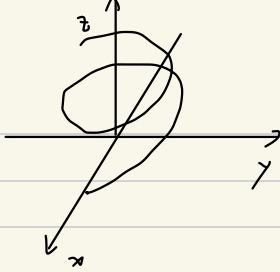
$$\begin{aligned}\int_C F \cdot dr &= \int_0^1 (2t^3, t+3t^2-6, t) \cdot (1, 2t, 3t^2) dt \\&= \int_0^1 2t^3 + 2t^2 + 6t^3 - 12t + 3t^2 dt \\&= -\frac{31}{12}\end{aligned}$$

1b $C: x = \cos t \quad y = \sin t \quad t \in (0, 2\pi)$

$$\begin{aligned}\int_C F \cdot dr &= \int_0^{2\pi} (0, \cos t + 3 \sin t, \cos t) \cdot (-\sin t, \cos t, 0) dt \\&= \int_0^{2\pi} \cos^2 t + 3 \sin t \cos t dt \\&= \int_0^{2\pi} \frac{\cos 2t + 1}{2} + \frac{3 \sin 2t}{2} dt \\&= \left(\frac{t}{2} + \frac{1}{4} \sin 2t - \frac{3}{4} \cos 2t \right)_0^{2\pi} = \pi\end{aligned}$$

2. $\int_C F \cdot dr = \int_0^{2\pi} (-\sin t, \cos t, t^2) \cdot (-\sin t, \cos t, 1) dt$

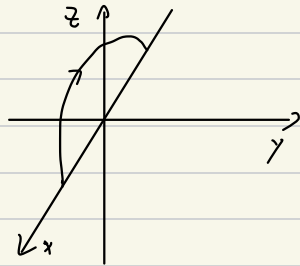
$$= \int_0^{2\pi} 1 + t^2 dt = 2\pi + \frac{8\pi^3}{3}$$



$$3. \text{ a) } \int_0^{\pi} (0, \cos t, \sin^2 t) (-\sin t, 0, \cos t) dt$$

$$= \int_0^{\pi} \sin^2 t \cos t dt$$

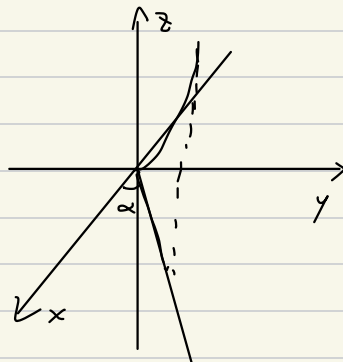
$$= \int_0^{\pi} \sin^2 t d \sin t = 0$$



$$\text{b) } \int_0^1 (-t \sin \alpha, t \cos \alpha, t^3) (\cos \alpha, \sin \alpha, 3t^2) dt$$

$$= \int_0^1 3t^5 dt$$

$$= \frac{1}{2}$$



$$\text{c) } \int_0^1 \left[-(1-t) \sin \alpha, (1-t) \cos \alpha, (1-t)^6 \right] \cdot \left[-\cos \alpha, -\sin \alpha, -3(1-t)^5 \right] dt$$

$$= -\frac{1}{2}$$

$$4. \quad 1a) \quad dr = (it, 2, 3t^2) dt$$

$$\int_0^1 2t^6 (2t, 2, 3t^2) dt = \int_0^1 (4t^7, 4t^6, 6t^8) dt$$

$$= \left(\frac{1}{2}, \frac{4}{7}, \frac{2}{3} \right)$$

$$1b) \quad \int_0^1 \begin{vmatrix} i & j & k \\ 2t^3 & 0 & t^4 \\ 2t & 2 & 3t^2 \end{vmatrix} dt = \int_0^1 (-2t^4, \cancel{4}t^5, 4t^3) dt$$

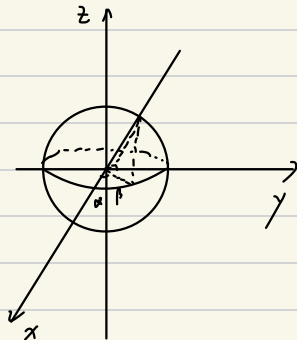
$$= \left(-\frac{2}{5}, -\frac{2}{3}, 1 \right)$$

Problem Sheet 3

$$1. (a) \int_{-1}^1 \int_{-1}^1 (0, 0, z) \cdot (0, 1, 0) dz dx$$

$$= 0$$

$$(b) \quad \begin{aligned} x &= \cos \alpha \cos \beta & \vec{n} &= (\cos \alpha \cos \beta, \sin \alpha \cos \beta, \sin \beta) \\ y &= \sin \alpha \cos \beta \\ z &= \sin \beta \end{aligned}$$



$$I = \int_0^{2\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (\sin \alpha \cos \alpha \cos^2 \beta, \sin^2 \alpha \cos^2 \beta, \sin \beta) \cdot (-\sin \alpha \cos \beta, \cos \alpha \cos \beta, 0) \times (-\cos \alpha \sin \beta, -\sin \alpha \sin \beta, \cos \beta) d\beta d\alpha$$

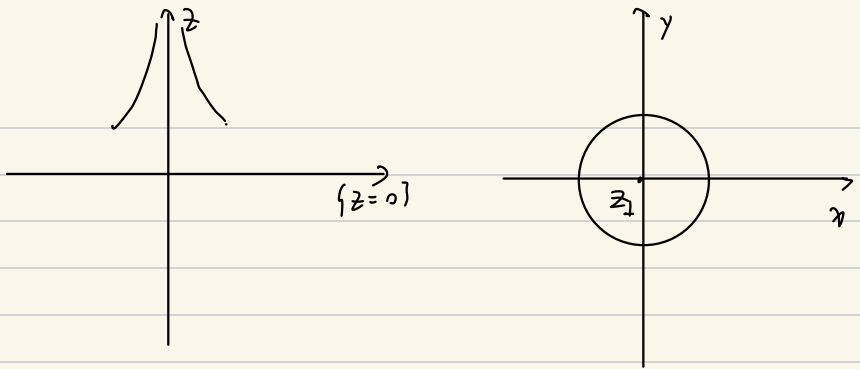
$$= \int_0^{2\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \begin{vmatrix} \sin \alpha \cos \alpha \cos^2 \beta & \sin^2 \alpha \cos^2 \beta & \sin \beta \\ -\sin \alpha \cos \beta & \cos \alpha \cos \beta & 0 \\ -\cos \alpha \sin \beta & -\sin \alpha \sin \beta & \cos \beta \end{vmatrix} d\beta d\alpha$$

$$= \int_0^{2\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin \beta \cdot (\sin^2 \alpha \sin \beta \cos \beta + \cos^2 \alpha \sin \beta \cos \beta) + \cos \beta (\sin \alpha \cos^2 \alpha \cos^3 \beta + \sin^3 \alpha \cos^3 \beta) d\beta d\alpha$$

$$= \int_0^{2\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} (\sin^4 \beta \cos \beta + \cos^4 \beta \sin \beta) d\beta d\alpha$$

$$= \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 2\pi \sin^2 \beta \cos \beta d\beta = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 2\pi \sin^2 \beta d\sin \beta = 2\pi \frac{1}{3} \sin^3 \beta \Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = \frac{4\pi}{3}$$

2 (a)



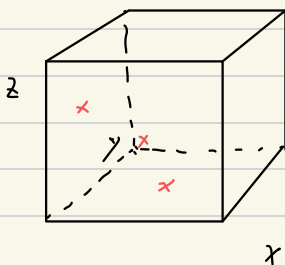
(b) $x = r \cos \theta$ $y = r \sin \theta$ $z = z$ $r > 0$, $\theta \in [0, 2\pi)$

$$\begin{aligned}
 V_0 &= \iiint_{\Omega} 1 \, dV = \iiint_{\Omega'} \begin{vmatrix} \cos \theta & \sin \theta & 0 \\ -r \sin \theta & r \cos \theta & 0 \\ 0 & 0 & 1 \end{vmatrix} dr \, d\theta \, dz \\
 &= \int_1^{+\infty} \int_0^{\frac{1}{z}} \int_0^{2\pi} r \, d\theta \, dr \, dz \\
 &= \int_1^{+\infty} \int_0^{\frac{1}{z}} 2\pi r \, dr \, dz \\
 &= \int_1^{+\infty} \pi r^2 \Big|_0^{\frac{1}{z}} dz = \int_1^{+\infty} \frac{\pi}{z^3} dz = -\frac{\pi}{z} \Big|_1^{+\infty} \\
 &= \pi
 \end{aligned}$$

(c) it is clear that area of S is larger than

that of $\{(x, y, z) \mid x^2 + y^2 = 1, z \geq 1\}$

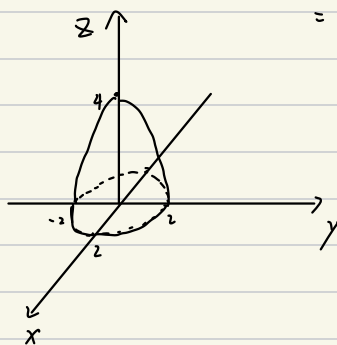
3,



$$\iint_S \mathbf{r} \cdot \mathbf{n} \, dS = 3 \iint_{S'} (1, y, z) \cdot (1, 0, 0) \, dS = 3$$

$$4. \iint_S (x, y, 1) \cdot (0, 0, 1) \, dS = \frac{1}{2}$$

$$\begin{aligned} 5. \iiint_V (1 + x^2 + y^2) \, dV &= \int_0^1 \int_0^1 \int_0^1 (1 + x^2 + y^2) \, dx \, dy \, dz \\ &= \int_0^1 \int_0^1 (1 + x^2 + y^2) \, dx \, dy \\ &= \int_0^1 \left(1 + \frac{1}{3} + y^2 \right) \, dy \\ &= \frac{5}{3} \end{aligned}$$



$$(b) \iiint_V ax + by + cz \, dV$$

$$x = r \cos \theta \quad y = r \sin \theta \quad z = z \quad r^2 + z \leq 4$$

$$I = \int_0^4 \int_0^{\sqrt{4-z}} \int_0^{2\pi} r (ar \cos \theta + br \sin \theta + cz) \, d\theta \, dr \, dz = \frac{32\pi c}{3}$$

$$7. I = \int_0^4 \int_0^{4-z} \int_0^{2\pi} \dots = \frac{4}{3} \pi c$$

Problem Sheet 4.

1. Verify Stokes $\oint_{\partial S} \mathbf{F} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{F}) \cdot \vec{n} \, dS$

(a) $x = 2 \cos t \quad y = 2 \sin t \quad z = 0$

$$\oint_{\partial S} \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} (-2 \sin t, 2 \cos t, 1) \times (-2 \sin t, 2 \cos t, 0) \, dt$$

$$= 8\pi$$

$$\iint_S (\nabla \times \mathbf{F}) \cdot \vec{n} \, dS = \iint_S \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -y & x & 1 \end{vmatrix} \cdot (0, 0, 1) \, dS$$

$$= \iint_S 1 \cdot dS = 8\pi$$

(b) $x = 2 \cos \theta \quad y = 2 \sin \theta \quad z = 2$

$$\oint_{\partial S} \mathbf{F} \cdot d\mathbf{r} = \int_0^{2\pi} (-2 \sin \theta, 2 \cos \theta, 1) \cdot (-2 \sin \theta, 2 \cos \theta, 0) \, d\theta = 8\pi$$

$x = 2 \cos \alpha \cos \beta \quad y = 2 \sin \alpha \cos \beta \quad z = 2 \sin \beta \quad \beta \in (0, \frac{\pi}{2})$

$$\iint_S (\nabla \times \mathbf{F}) \cdot \vec{n} \, dS = \iint_S (0, 0, 2) \cdot \vec{n} \, dS$$

$$= \int_0^{2\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \begin{vmatrix} 0 & 0 & 2 \\ -2 \sin \alpha \cos \beta & 2 \cos \alpha \cos \beta & * \\ -2 \cos \alpha \sin \beta & -2 \sin \alpha \sin \beta & * \end{vmatrix} \, d\alpha \, d\beta$$

$$= \int_0^{2\pi} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} 2 \cdot (4 \sin^2 \alpha \cos \beta \sin \beta + 4 \cos^2 \alpha \sin \beta \cos \beta) \, d\alpha \, d\beta$$

$$= 8\pi \int_0^{\frac{\pi}{2}} \sin 2\beta \, d\beta = 8\pi$$

$$\begin{aligned}
2 \quad \oint_C \mathbf{F} \cdot d\mathbf{r} &= \iint_S (\nabla \times \mathbf{F}) \cdot \vec{n} \, ds \quad S = \{ (x, y, z) \mid z=0, x^2+y^2 \leq 1 \} \\
&= \iint_S \begin{pmatrix} 2x^2 + 4x^2y e^{xy^2} - 0 \\ 0 - 4xy - 4xe^{xy^2} - 2xy^2 e^{xy^2} \\ -2y^2 \end{pmatrix} \cdot (0, 0, 1) \, ds \\
&= \iint_S -2y^2 \, ds = \iint_S -(x^2+y^2) \, ds = \int_0^1 -r \cdot 2\pi \cdot dr \\
&= -\frac{\pi}{2}
\end{aligned}$$

Problem Sheet 5.

1. (a)
$$\iiint_V \nabla \cdot \mathbf{F} \, dV = \iint_{\partial V} \mathbf{F} \cdot \vec{n} \, dS$$

(b) $S: F(x, y, z) = 1 \quad n_s = \nabla F = \frac{(2x, 2y, 2z)}{\sqrt{(2x)^2 + (2y)^2 + (2z)^2}} = (x, y, z)$

(c) $a \cdot (x, y, z) = x^4 + y^4 + z^4$
 $a = (x^3, y^3, z^3)$ satisfies.

2.
$$\iiint_V \nabla \cdot \mathbf{F} \, dV = \iint_{\partial V} \mathbf{F} \cdot \vec{n} \, dS$$

$x = \cos \alpha \cos \beta$
 $y = \sin \alpha \cos \beta$
 $z = \sin \beta$
 $\alpha \in (0, 2\pi)$
 $\beta \in (-\frac{\pi}{2}, \frac{\pi}{2}) = 2\pi$

$$= \iint_{S^2} (0, 0, \frac{z^{13}}{[1+(2x^2+y^2)]^{1/2}}) \cdot \vec{n} \, dS$$

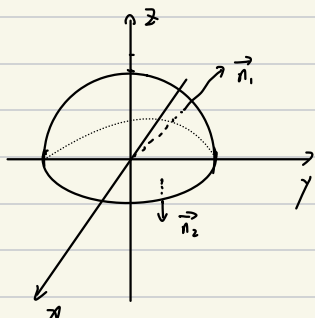
$$= \int_0^{2\pi} d\alpha \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \begin{vmatrix} 0 & 0 & \frac{\sin^{13} \beta}{2^{2010}} \\ -\sin \alpha \cos \beta & \cos \alpha \cos \beta & 0 \\ -\cos \alpha \sin \beta & -\sin \alpha \sin \beta & \cos \beta \end{vmatrix} d\beta$$

$$= 2\pi \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sin^{13} \beta}{2^{2010}} \sin \beta \cos \beta \, d\beta$$

$$= \frac{\pi}{2^{2009}} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^{14} \beta \, d\sin \beta = \frac{\left[\frac{1}{15} \sin^{15} \beta \right]_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \pi}{2^{2009}}$$

$$= \frac{\pi}{15 \cdot 2^{2009}}$$

3. (a)



(b)
$$\oint_S \mathbf{F} \cdot \vec{n} \, dS = \iiint_V \nabla \cdot \mathbf{F} \, dV$$

$$\begin{aligned}
 (C) \quad I &= \iiint_V \cancel{3x^2} + (-\cancel{3x^2}) + 2x^2 z \, dV \\
 &= \iiint_V 2x^2 z \, dV \\
 &= \int_0^3 \int_0^{\sqrt{9-z^2}} \int_0^{2\pi} 2r^2 \cos^2 \theta \cdot z \cdot r \, d\theta \, dr \, dz \\
 &= \int_0^3 \int_0^{\sqrt{9-z^2}} \int_0^{2\pi} (\cos 2\theta + 1) r^3 z \, d\theta \, dr \, dz \\
 &= \int_0^3 \int_0^{\sqrt{9-z^2}} 2\pi r^3 z \, dr \, dz = \int_0^3 2\pi \frac{1}{4} (9-z^2)^2 \cdot z \, dz \\
 &= -\frac{\pi}{4} \int_0^3 (9-z^2)^2 d(9-z^2) \\
 &= \frac{1}{3} \cdot \left(-\frac{\pi}{4}\right) \cdot (9-z^2)^3 \Big|_0^3 \\
 &= \frac{3^5}{4} \pi
 \end{aligned}
 \quad \begin{cases} x = r \cos \theta \\ y = r \sin \theta \\ z = z \end{cases}$$

$$\begin{aligned}
 1. (d) \quad I &= \oint_S \mathbf{a} \cdot \vec{n} \, dS = \iiint_V (\nabla \cdot \mathbf{a}) \, dV \\
 &= \iiint_{p^3} 3(x^2 + y^2 + z^2) \, dV \\
 &= \int_0^1 3 \cdot r^2 \cdot 4\pi r^2 \, dr = \frac{12\pi}{5}
 \end{aligned}$$

Problem Sheet 6.

$$1. \quad (a) \quad (a \cdot c)(b \cdot d) \quad (b) \quad b \cdot c \\ (c) \quad b \times a \quad (d) \quad a \cdot (c \times b)$$

$$2. \quad (a) \quad 3 \quad (b) \quad \delta_{ik} \quad (c) \quad 3 \quad (d) \quad 0$$

$$3. \quad \epsilon_{ijk} \epsilon_{ilm} = \epsilon_{jki} \epsilon_{ilm} = \delta_{jl} \delta_{km} - \delta_{jm} \delta_{kl}$$

$$\epsilon_{ijk} \epsilon_{ijm} = 3 \delta_{km} - \delta_{jm} \delta_{kj} = 2 \delta_{km}$$

$$\epsilon_{ijk} \epsilon_{ijl} = 6$$

$$4. \quad a \cdot (b \times c) = a_i \epsilon_{ijk} b_j c_k = b_j \epsilon_{jki} c_k a_i = b \cdot (c \times a)$$

$$\begin{aligned} 5. & \quad e_i \epsilon_{ijk} a_j b_k \cdot e_l \epsilon_{lmn} c_m d_n \\ & = \epsilon_{ijk} \epsilon_{lmn} a_j b_k c_m d_n \\ & = (\delta_{jm} \delta_{kn} - \delta_{jn} \delta_{km}) a_j b_k c_m d_n = a_j b_k c_j d_k - a_j b_k c_n d_j \\ & = (a \cdot c)(b \cdot d) - (a \cdot d)(b \cdot c) \end{aligned}$$

$$6.1a \quad \begin{vmatrix} a_1 & a_2 & a_3 \\ a_1 & a_2 & a_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = 0 \quad \text{by definition of Det}(\cdot)$$

$$(b) \quad (a+b) \cdot (b+c) \times (c+a) = \det \begin{pmatrix} a+b \\ b+c \\ c+a \end{pmatrix} = \det \begin{pmatrix} a \\ b \\ c \end{pmatrix} = 2 a \cdot (b \times c)$$

Problem Sheet. 7.

$$(\psi_1 \partial_1 \phi, \psi_2 \partial_2 \phi, \psi_3 \partial_3 \phi)$$

$$1. \nabla \cdot (\psi \nabla \phi) = \partial_i (\psi \partial_i \phi_i) = \partial_i \psi \partial_i \phi_i + \psi \partial_i^2 \phi_i = \nabla \psi \cdot \nabla \phi + \psi \nabla^2 \phi$$

$$\begin{aligned} \nabla \times (\psi \nabla \phi) &= \epsilon_{ijk} e_i \partial_j (\psi \partial_k \phi_k) = \epsilon_{ijk} e_i (\partial_j \psi \partial_k \phi_k + \psi \partial_j \partial_k \phi_k) \\ &= e_i \epsilon_{ijk} (\partial_j \psi \partial_k \phi_k + \psi \partial_j \partial_k \phi_k) \\ &= \nabla \psi \times \nabla \phi \end{aligned}$$

$$\begin{aligned} 2. \nabla \times (a \times r) &= \epsilon_{ijk} e_i \partial_j (a \times r)_k \\ &= \epsilon_{ijk} e_i \partial_j \cdot \delta_{kl} \cdot a_m \cdot r_n \cdot \epsilon_{lmn} \\ &= \epsilon_{ijk} \epsilon_{kmn} e_i a_m \partial_j r_n \\ &= (\delta_{in} \delta_{jm} - \delta_{im} \delta_{jn}) e_i a_m \partial_j r_n \\ &= e_i a_i \partial_j r_j - e_i a_j \partial_j r_i \\ &= a(\nabla \cdot r) - a \cdot \nabla r = 2a \end{aligned}$$

$$\begin{aligned} 3. \nabla \times (\vec{r}/r^n) &= \epsilon_{ijk} e_i \frac{\partial}{\partial x_j} \left(\frac{\vec{r}}{r^n} \right)_k \\ &= \epsilon_{ijk} e_i \frac{\partial}{\partial x_j} \frac{x_k}{r^n} \\ &= \epsilon_{ijk} e_i \left(\frac{\partial x_k}{\partial x_j} \cdot \frac{1}{r^n} - x_k \cdot \frac{n}{r^{n+1}} \cdot \frac{\partial r}{\partial x_j} \right) \end{aligned}$$

$$r^2 = x_m^2$$

$$2r \cdot \frac{\partial r}{\partial x_j} = 2x_m \cdot \frac{\partial x_m}{\partial x_j} = 2x_m \delta_{mj} = 2x_j$$

$$\begin{aligned} &= e_i \epsilon_{ijk} \frac{1}{r^n} \delta_{jk} - \frac{n}{r^{n+2}} e_i \epsilon_{ijk} x_j x_k \\ &= 0 \end{aligned}$$

$$\nabla \cdot (\vec{r}/r^n) = \frac{\partial}{\partial x_j} \frac{x_j}{r^n} = \frac{\delta_{jj}}{r^n} - \frac{n x_i x_i}{r^{n+2}} = (3-n) r^{-n}$$

Problem Sheet. 8.

$$1. (a) \quad h_u = \frac{\partial \mathbf{r}}{\partial u} = 1 \quad e_u = (1, 0, 0) \\ h_v = v \quad e_v = (0, 1, 0) \\ h_w = 3w^2 \quad e_w = (0, 0, 1)$$

$$(b) \quad e_u \cdot e_v = 0 \quad \dots$$

$$(c) \quad dV = 6vw^2 \, du \, dv \, dw$$

$$(d) \quad \nabla f = \frac{\partial f}{\partial x} e_x + \frac{\partial f}{\partial y} e_y + \frac{\partial f}{\partial z} e_z \\ = \frac{\partial f}{\partial u} e_u + \frac{\partial f}{\partial v} \cdot \frac{\partial v}{\partial u} e_u + \frac{\partial f}{\partial z} \cdot \frac{\partial z}{\partial w} e_w$$

$$2. \quad \mathbf{r} = (x, y, z) = (a \cosh u \cos v, a \sinh u \sin v, w)$$

$$h_u = \left| \frac{\partial \mathbf{r}}{\partial u} \right| = \left| \left(a \sinh u \cos v, a \cosh u \sin v, 0 \right) \right| = a \sqrt{\frac{e^{2u} + e^{-2u}}{4} \cos^2 v + \frac{e^{-2u} + e^{2u}}{4} \sin^2 v} \\ = a \left[\frac{1}{2} \cosh 2u + \frac{1}{2} \cos 2v \right]^{\frac{1}{2}}$$

⋮

$$\nabla f = \frac{\partial f}{\partial u} \cdot \frac{1}{h_u} \cdot e_u + \frac{\partial f}{\partial v} \frac{1}{h_v} e_v + \frac{\partial f}{\partial w} \frac{1}{h_w} e_w$$

$$3. (a) \quad \nabla^2 f = \nabla \cdot (2x, 2y, 2z) = 6$$

$$(b) \quad x = r \cos \theta \quad y = r \sin \theta \quad z = z$$

$$h_1 = \left| \frac{\partial \vec{r}}{\partial r} \right| = \left| \frac{\partial (r \cos \theta, r \sin \theta, z)}{\partial r} \right| = |(\cos \theta, \sin \theta, 0)| = 1$$

$$h_2 = \left| \frac{\partial \vec{r}}{\partial \theta} \right| = |(-r \sin \theta, r \cos \theta, 0)| = r$$

$$h_3 = \left| \frac{\partial \vec{r}}{\partial z} \right| = |(0, 0, 1)| = 1$$

$$\begin{aligned}\nabla^2 f &= \frac{1}{r} \left[\frac{\partial}{\partial r} (r (2r)) + \frac{\partial}{\partial \theta} \left(\frac{1}{r} \cdot 0 \right) + \frac{\partial}{\partial z} (r \cdot 2z) \right] \\ &= \frac{1}{r} (4r + 0 + 2r) = 6\end{aligned}$$

$$(c) \quad x = r \cos \alpha \cos \beta \quad y = r \sin \alpha \cos \beta \quad z = r \sin \beta$$

$$h_1 = \left| \frac{\partial \vec{r}}{\partial r} \right| = \left| \frac{\partial (r \cos \alpha \cos \beta, r \sin \alpha \cos \beta, r \sin \beta)}{\partial r} \right| = \left| (\cos \alpha \cos \beta, \sin \alpha \cos \beta, \sin \beta) \right| = 1$$

$$h_2 = \left| (-r \sin \alpha \cos \beta, r \cos \alpha \cos \beta, 0) \right| = r \cos \beta$$

$$h_3 = \left| (-r \cos \alpha \sin \beta, -r \sin \alpha \sin \beta, r \cos \beta) \right| = r$$

$$\begin{aligned}\nabla^2 f &= \frac{1}{r^2 \cos \beta} \left[\frac{\partial}{\partial r} (r^2 \cos \beta \cdot 2r) + 0 + 0 \right] \\ &= \frac{6r^2 \cos \beta}{r^2 \cos \beta} = 6\end{aligned}$$

$$4. \quad x = r \cos \alpha \cos \beta \quad y = r \sin \alpha \cos \beta \quad z = r \sin \beta$$

$$\vec{e}_r = \frac{\partial (r \cos \alpha \cos \beta, r \sin \alpha \cos \beta, r \sin \beta)}{\partial r} = (\cos \alpha \cos \beta, \sin \alpha \cos \beta, \sin \beta)$$

$$\vec{e}_\alpha = (-r \sin \alpha \cos \beta, r \cos \alpha \cos \beta, 0)$$

$$\vec{e}_\beta = (-r \cos \alpha \sin \beta, -r \sin \alpha \sin \beta, r \cos \beta)$$

$$\nabla \cdot \vec{e}_r = \frac{1}{1 \cdot r \cos \beta \cdot r} \left[\frac{\partial}{\partial r} (r^2 \cos \beta) \right]$$

$$= \frac{2}{r}$$

$$\nabla \cdot \vec{e}_\alpha = \frac{1}{1 \cdot r \cos \beta \cdot r} \left(\frac{\partial}{\partial \alpha} \cdot r \right) = 0 \quad \checkmark$$

$$\nabla \cdot \vec{e}_\beta = \frac{1}{1 \cdot r \cos \beta \cdot r} \left(\frac{\partial}{\partial \beta} \cdot r \cos \beta \right) = -\frac{\tan \beta}{r}$$

Proof of curvilinear coordinates.

$$1. \quad df = \nabla f \cdot dr$$

$$\begin{cases} df = \sum_i \frac{\partial f}{\partial u_i} du_i \\ \nabla f = \sum_i a_i e_i \\ dr = \sum_i e_i du_i \end{cases}$$

$$\nabla f \cdot dr = \sum_{i,j} a_i (e_i \cdot e_j) du_j = \sum_i \frac{\partial f}{\partial u_i} du_i$$

$$\text{then } \frac{\partial f}{\partial u_i} = \sum_j (e_i \cdot e_j) a_j$$

$$\underbrace{\begin{pmatrix} e_1 \cdot e_1 & \dots & e_1 \cdot e_n \\ e_2 \cdot e_1 & \dots & e_2 \cdot e_n \\ \vdots & \ddots & \vdots \\ e_n \cdot e_1 & \dots & e_n \cdot e_n \end{pmatrix}}_G \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} = \begin{pmatrix} \frac{\partial f}{\partial u_1} \\ \vdots \\ \frac{\partial f}{\partial u_n} \end{pmatrix}$$

$$\vec{a} = \frac{\partial f}{\partial u} \cdot G^{-1}$$

$$\nabla f = \sum_{i,j} g^{ij} \frac{\partial f}{\partial u_j} e_i$$

$$\text{if orthogonal } g_{ij} = h_i^2$$

$$\nabla f = \sum_i \frac{1}{h_i^2} e_i \cdot \frac{\partial f}{\partial u_i} = \sum_i \frac{\hat{e}_i}{h_i} \frac{\partial f}{\partial u_i}$$

$$2. \quad \nabla \cdot A = \lim_{V \rightarrow 0} \frac{1}{V} \iint_{\partial V} A \cdot \vec{n} \, dS$$

$$\iint_{\partial V} A \cdot \vec{n} \, dS$$

$$= \left[\left(\iint_{S_1} + \iint_{S_2} \right) + \left(\iint_{S_3} + \iint_{S_4} \right) + \left(\iint_{S_5} + \iint_{S_6} \right) \right]$$

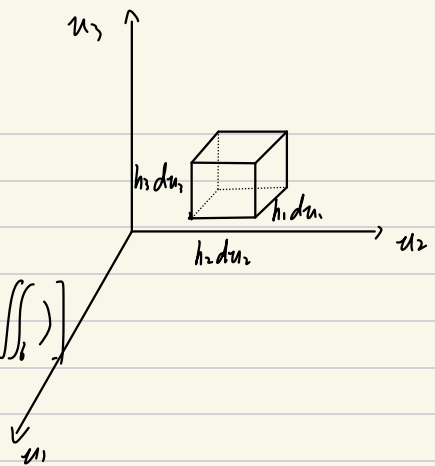
$$(A \cdot \vec{n} \, dS)$$

$$= A(u_1 + du_1, u_2, u_3) \cdot (1, 0, 0) \cdot h_2 h_3 \, du_2 \, du_3 - A(u_1, u_2, u_3) \cdot (1, 0, 0) h_2 h_3 \, du_2 \, du_3$$

+ - - - -

$$= \frac{\partial}{\partial u_1} (A h_2 h_3) \, du_1 \, du_2 \, du_3 + \dots = \frac{1}{h_1 h_2 h_3} \sum_i \frac{\partial}{\partial u_i} \left(\frac{\pi h_i}{h_i} A_i \right)$$

$$3. \quad \nabla^2 f = \frac{1}{h_1 h_2 h_3} \sum \frac{\partial}{\partial u_i} \left(\frac{\pi h_i}{h_i^2} \cdot \frac{\partial f}{\partial u_i} \right)$$



$$\oint (\nabla \times A) \cdot \vec{n} = \lim_{\delta \rightarrow 0} \frac{1}{\delta} \cdot \oint_{\partial S} A \cdot d\vec{r}$$

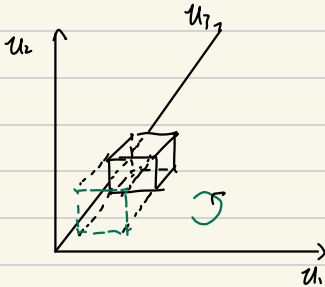
$$(u_1, u_2, u_3)$$

$$A = \sum_i^3 A_i \hat{e}_i$$

$$\text{fix } u_3$$

$$dS = h_1 h_2 du_1 du_2$$

$$\vec{n} = \hat{e}_3$$



$$\begin{aligned} \oint_{\partial S} A \cdot d\vec{r} &= \left(\oint_{e_1} + \oint_{e_2} + \oint_{e_3} + \oint_{e_4} \right) (A \cdot d\vec{r}) \\ &= (\oint_{e_1} + \oint_{e_3}) (A \cdot d\vec{r}) + (\oint_{e_2} + \oint_{e_4}) (A \cdot d\vec{r}) \\ &\equiv - \frac{\partial}{\partial u_2} (A_1 h_1) du_1 du_2 + \frac{\partial}{\partial u_1} (A_2 h_2) du_1 du_2 \end{aligned}$$

$$\text{similarly fix } u_2, u_1$$

$$(\nabla \times A)_i = \frac{1}{h_j h_k} \left[\frac{\partial}{\partial u_j} (A_k h_k) - \frac{\partial}{\partial u_k} (A_j h_j) \right]$$

$$\nabla \times A = \frac{1}{h_1 h_2 h_3} \begin{vmatrix} h_1 \hat{e}_1 & h_2 \hat{e}_2 & h_3 \hat{e}_3 \\ \frac{\partial}{\partial u_1} & \frac{\partial}{\partial u_2} & \frac{\partial}{\partial u_3} \\ A_1 h_1 & A_2 h_2 & A_3 h_3 \end{vmatrix}$$